

Problem

Use Laplace transforms to solve for $x(t)$ in

$$x(t) = \cos t + \int_0^t e^{\lambda-t} x(\lambda) d\lambda$$

Step-by-step solution

Step 1 of 4

Consider the following equation:

$$x(t) = \cos t + \int_0^t e^{\lambda-t} x(\lambda) d\lambda \quad \dots\dots (1)$$

Step 2 of 4

We notice from equation (1) the term $\int_0^t e^{\lambda-t} x(\lambda) d\lambda$ is in convolution integral form.

Define convolution integral.

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \int_0^t x(\lambda) h(t-\lambda) d\lambda \end{aligned}$$

Now assume impulse response $h(t)$ is e^{-t}

Now, convolution integral becomes:

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \int_0^t x(\lambda) h(t-\lambda) d\lambda \\ &= \int_0^t x(\lambda) e^{-(t-\lambda)} d\lambda \\ &= \int_0^t x(\lambda) e^{\lambda-t} d\lambda \end{aligned}$$

Above expression exactly matches with term $\int_0^t e^{\lambda-t} x(\lambda) d\lambda$.

Thus our assumption impulse response $h(t) = e^{-t}$ is valid.

Step 3 of 4

We know the convolution integral relates the convolution of two signals in the time domain to the inverse of the product of their Laplace transforms:

Rewrite the equation (1).

$$x(t) = \cos t + e^{-t} * x(t)$$

Write the Laplace transform of each term in the above expression.

$$L[x(t)] = L[\cos t] + L[e^{-t}]L[x(t)]$$

$$X(s) = \frac{s}{s^2 + 1} + \frac{1}{s + 1}(X(s)) \quad \text{since} \quad \begin{cases} e^{-at} = \frac{1}{s + a} \text{ and} \\ \cos \omega t = \frac{s}{s^2 + \omega^2} \end{cases}$$

Rewrite the expression.

$$X(s) \left[1 - \frac{1}{s + 1} \right] = \frac{s}{s^2 + 1}$$

$$X(s) \left(\frac{s}{s + 1} \right) = \frac{s}{s^2 + 1}$$

$$X(s) = \frac{s + 1}{s^2 + 1}$$

Thus,

$$X(s) = \frac{s}{s^2 + 1} + \frac{1}{s^2 + 1}$$

Take inverse Laplace transform to the above expression.

$$L^{-1}[X(s)] = L^{-1} \left[\frac{s}{s^2 + 1} \right] + L^{-1} \left[\frac{1}{s^2 + 1} \right]$$

$$x(t) = \cos t + \sin t \quad \text{since} \quad \begin{cases} \cos \omega t = \frac{s}{s^2 + \omega^2} \text{ and} \\ \sin \omega t = \frac{\omega}{s^2 + \omega^2} \end{cases}$$

Step 4 of 4

The above expression can be rewrite as:

$$x(t) = \sqrt{2} \cos t \left(\frac{1}{\sqrt{2}} \right) + \sqrt{2} \sin t \left(\frac{1}{\sqrt{2}} \right)$$

$$= \sqrt{2} \left(\cos t \cos 45^\circ + \sin t \sin 45^\circ \right) \quad \text{since} \quad \begin{cases} \cos 45^\circ = \frac{1}{\sqrt{2}} \text{ and} \\ \sin 45^\circ = \frac{1}{\sqrt{2}} \end{cases}$$

$$= \sqrt{2} \cos(t - 45^\circ) \quad \text{since} \quad \begin{cases} \cos(A - B) \\ = \cos A \cos B + \sin A \sin B \end{cases}$$

Therefore $x(t)$ is $\boxed{\sqrt{2} \cos(t - 45^\circ)}$.